# The stationary-start maintenance model and its exact delay-period predictions

Working note

September 7, 2026

**Abstract**

One change to the maintenance model — initialising the Ornstein–Uhlenbeck maintenance noise at its stationary law rather than at zero — makes the transient irrelevant to the delay-period analysis window, and every prediction the paper tests then follows from the *stationary* Kac–Rice rate by a short corollary. The note is organised accordingly: Sections 1–4 establish that the transient can be neglected, and state precisely under what conditions; Sections 5–7 work with the stationary rate alone and derive the predictions. Every statement has been verified numerically (Section 9).

## Contents

**1 The model** **1**

**2 Exact second-order moments** **2**

**3 How fast the transient dies** **3**

**4 Neglecting the transient** **4**
4.1 The three conditions . . . . . 4
4.2 Proof of Theorem 8 . . . . . 5

**5 The stationary rate** **6**

**6 The standardised distance is measured** **7**

**7 Predictions** **7**

**8 Summary** **8**

**9 Numerical verification** **8**

**10 Consequences for the paper** **8**

## 1 The model

Everything before stimulus offset is unchanged. Maintenance begins at $\tau = t_{\text{offset}} + L_{\text{motion}}$, at which time the gate-facing signal $y$ equals the maintained target $m$. During maintenance,

$$\mathrm{d}\varepsilon(t) = -\kappa\,\varepsilon(t)\,\mathrm{d}t + \sigma_\varepsilon\,\mathrm{d}W_\varepsilon(t), \tag{1}$$

$$\mathrm{d}y(t) = \lambda\,[\,m + \varepsilon(t) - y(t)\,]\,\mathrm{d}t, \tag{2}$$

and $H := y - m$ obeys $\mathrm{d}H = \lambda(\varepsilon - H)\,\mathrm{d}t$.

**Initial conditions.** $H(\tau) = 0$ is *forced*: the filter output is continuous and $m$ is defined as the value of $y$ at $\tau$. It is not a modelling choice. By contrast $\varepsilon(\tau)$ is free; the present model sets it to 0. We propose instead

$$\varepsilon(\tau) \sim \mathrm{N}\big(0,\ \sigma_\varepsilon^2/2\kappa\big) \quad \text{independent of } W_\varepsilon \tag{ss}$$

— the stationary law of (1). Call (ss) the *stationary start* and $\varepsilon(\tau) = 0$ the *cold start*.

**Why.** $\varepsilon$ represents fluctuations intrinsic to holding a value. Nothing makes such a process sit exactly at its mean at the instant maintenance begins, and no measurement constrains it. As Corollary 6 shows, the cold start is the sole source of a slow $(1/\kappa)$ transient in a system whose only other timescale is the fast filter $1/\lambda$; under (ss) the model's delay-period predictions become exact rather than asymptotic.

*Remark* 1. (ss) removes the mechanism that gives Model 1 its only delay-period ramp. That is not a way of disarming Model 1: Section 4 bounds what *any* residual non-stationarity can buy it, and the bound is far below the data.

**Notation.** $\rho := t - \tau$ is elapsed maintenance time; $\beta/\delta$ is the speed of temporal preparation, with $\beta = 0$ under Model 1; $d(t) = g - |m| - (\beta/\delta)(t - t_{\text{offset}})^+$ is the distance to the gate; $\varphi, \Phi$ are the standard normal density and c.d.f., and $g(u) := u\Phi(u) + \varphi(u)$, so $g' = \Phi$ and $g'' = \varphi$. Write $s^2 := \sigma_\varepsilon^2/2\kappa$ and

$$A_\infty := \frac{\lambda s^2}{\lambda + \kappa}, \qquad B_\infty := \frac{\lambda^2 \sigma_\varepsilon^2}{2(\lambda + \kappa)}, \qquad \omega := \sqrt{B_\infty/A_\infty} = \sqrt{\lambda\kappa}, \qquad a_\infty := \frac{d}{\sqrt{A_\infty}}. \tag{3}$$

## 2 Exact second-order moments

**Proposition 2** (exact moments)**.** *Under* (1)–(ss) *with* $H(\tau) = 0$*, put*

$$X(\rho) := \frac{(\lambda + \kappa)\big(e^{-2\lambda\rho} - e^{-(\lambda+\kappa)\rho}\big)}{\lambda - \kappa} \qquad \big(= -2\lambda\rho\, e^{-2\lambda\rho} \quad \textit{when } \lambda = \kappa\big). \tag{4}$$

*Then for every* $\rho > 0$*,*

$$\boxed{\frac{\sigma_H^2}{A_\infty} = 1 + X - e^{-(\lambda+\kappa)\rho}, \qquad \frac{c_H}{A_\infty} = -\lambda X, \qquad \frac{\sigma_{\dot H}^2}{B_\infty} = 1 + \frac{\lambda}{\kappa}\Big(X + e^{-(\lambda+\kappa)\rho}\Big).} \tag{5}$$

*In particular* $\sigma_H^2(0) = 0$*,* $c_H(0) = 0$*,* $\sigma_{\dot H}^2(0) = \frac{\lambda+\kappa}{\kappa} B_\infty$*, and all three converge to* (3) *with* $c_H(\infty) = 0$*.*

*Proof.* $(\varepsilon, H)$ is a two-dimensional linear diffusion with $F = \left(\begin{smallmatrix} -\kappa & 0 \\ \lambda & -\lambda \end{smallmatrix}\right)$ and $Q = \mathrm{diag}(\sigma_\varepsilon^2, 0)$, so its covariance solves $\dot P = FP + PF^\top + Q$ with $P(0) = \mathrm{diag}(s^2, 0)$. The system is triangular:

$$\dot P_{\varepsilon\varepsilon} = -2\kappa P_{\varepsilon\varepsilon} + \sigma_\varepsilon^2, \qquad P_{\varepsilon\varepsilon}(0) = s^2, \tag{6}$$

$$\dot P_{\varepsilon H} = -(\kappa + \lambda) P_{\varepsilon H} + \lambda P_{\varepsilon\varepsilon}, \qquad P_{\varepsilon H}(0) = 0, \tag{7}$$

$$\dot P_{HH} = 2\lambda(P_{\varepsilon H} - P_{HH}), \qquad P_{HH}(0) = 0. \tag{8}$$

By (6), $\dot P_{\varepsilon\varepsilon}(0) = -2\kappa s^2 + \sigma_\varepsilon^2 = 0$, so $P_{\varepsilon\varepsilon} \equiv s^2$: *the stationary start freezes the first component exactly.* Then (7) gives $P_{\varepsilon H} = A_\infty\big(1 - e^{-(\lambda+\kappa)\rho}\big)$, and (8), integrated from 0,

$$P_{HH} = 2\lambda \int_0^\rho e^{-2\lambda(\rho - v)} P_{\varepsilon H}(v)\mathrm{d}v = A_\infty \left[1 - e^{-2\lambda\rho} - 2\lambda \frac{e^{-(\lambda+\kappa)\rho} - e^{-2\lambda\rho}}{\lambda - \kappa}\right],$$

and writing $-2\lambda = -(\lambda+\kappa) - (\lambda-\kappa)$ in the numerator gives $P_{HH}/A_\infty - 1 = X - e^{-(\lambda+\kappa)\rho}$. Since $\dot{H} = \lambda(\varepsilon - H)$, $c_H = \lambda(P_{\varepsilon H} - P_{HH}) = -\lambda A_\infty X$ and $\sigma_{\dot{H}}^2 = \lambda^2(P_{\varepsilon\varepsilon} - 2P_{\varepsilon H} + P_{HH}) = B_\infty + \lambda^2 A_\infty\big(X + e^{-(\lambda+\kappa)\rho}\big)$, using $\lambda^2(s^2 - A_\infty) = B_\infty$ and $\lambda^2 A_\infty/B_\infty = \lambda/\kappa$. The $\lambda = \kappa$ forms follow by l'Hôpital. □

*Remark* 3. $\sigma_H$ rises from 0; $\sigma_{\dot{H}}$ *falls* from $\frac{\lambda+\kappa}{\kappa}B_\infty$, because $\dot{H}(\tau) = \lambda\varepsilon(\tau)$ has the noise at full amplitude while $H$ is still at zero. The position variance fills in while the velocity variance settles down.

## 3 How fast the transient dies

**Lemma 4.** *For $A, B > 0$ and $\rho > 0$, $\left|e^{-A\rho} - e^{-B\rho}\right| \le |A - B|\,\rho\, e^{-\min(A,B)\rho}$.*

*Proof.* The mean value theorem applied to $x \mapsto e^{-x\rho}$ on the interval with endpoints $A, B$ gives $|A - B|\rho e^{-\xi\rho}$ for some $\xi$ between them, and $e^{-\xi\rho} \le e^{-\min(A,B)\rho}$. □

**Proposition 5** (transient bounds)**.** *Set $\mu := \lambda + \min(\lambda, \kappa)$, $E_1 := e^{-(\lambda+\kappa)\rho}$, $E_2 := e^{-2\lambda\rho}$, and*

$$\mathcal{D}(\rho) := \min\left\{\rho\, e^{-\mu\rho},\ \frac{E_1 + E_2}{|\lambda - \kappa|}\right\} \qquad (\textit{first entry only, when } \lambda = \kappa). \tag{9}$$

*Then, for all $\lambda, \kappa, \rho > 0$,*

$$\bar{\Delta} := (\lambda+\kappa)\mathcal{D} + E_1, \qquad \bar{\gamma} := \lambda(\lambda+\kappa)\mathcal{D}, \qquad \bar{e} := \min\left\{\frac{\lambda}{\kappa}\bar{\Delta},\ \frac{\lambda\big[(\lambda+\kappa)E_2 + 2\kappa E_1\big]}{\kappa|\lambda - \kappa|}\right\} \tag{10}$$

*bound the three deviations:*

$$\left|\frac{\sigma_H^2}{A_\infty} - 1\right| \le \bar{\Delta}, \qquad 0 \le \frac{c_H}{A_\infty} \le \bar{\gamma}, \qquad \left|\frac{\sigma_{\dot{H}}^2}{B_\infty} - 1\right| \le \bar{e}. \tag{11}$$

*All three are decreasing on $\rho \ge 1$ when $\lambda \ge 1$.*

*Proof.* Everything reduces to bounding $|E_1 - E_2|/|\lambda - \kappa|$, since by Proposition 2 $X = (\lambda + \kappa)(E_2 - E_1)/(\lambda - \kappa)$, $\sigma_H^2/A_\infty - 1 = X - E_1$, $c_H/A_\infty = -\lambda X$, and $\sigma_{\dot{H}}^2/B_\infty - 1 = (\lambda/\kappa)\big[(\lambda + \kappa)E_2 - 2\kappa E_1\big]/(\lambda - \kappa)$, this last by adding and subtracting $(\lambda+\kappa)E_1$ in the numerator. Two bounds are available and we take the smaller. *(a)* By Lemma 4 with $A = 2\lambda$, $B = \lambda + \kappa$ — so $|A - B| = |\lambda - \kappa|$ and $\min(A, B) = \mu$ — $|E_1 - E_2|/|\lambda - \kappa| \le \rho e^{-\mu\rho}$. *(b)* Trivially $|E_1 - E_2| \le E_1 + E_2$. Bound (a) is the sharp one when $\lambda \approx \kappa$, where (b) is singular; (b) is the sharp one when the two rates are far apart, where (a)'s worst-case exponent $\mu$ is pessimistic. The sign $X \le 0$ follows by inspection of (4), giving $c_H \ge 0$. For the third bound, the first entry of $\bar{e}$ comes from the $X$ form and the second from bounding the two exponentials separately. Monotonicity: for the first entry, $\frac{\mathrm{d}}{\mathrm{d}\rho}(1 + c\rho)e^{-\mu\rho} = e^{-\mu\rho}(c - \mu - \mu c\rho)$ with $c = \lambda + \kappa \ge \mu$, negative for $\rho > (c - \mu)/(\mu c)$, and $(c - \mu)/(\mu c) \le 1/(2\lambda) \le 1/2$ when $\lambda \ge 1$; the second entry is a sum of decaying exponentials. □

**Corollary 6** (relaxation is at least as fast as $\lambda$)**.** From Propositions 2 and 5. *All three moments converge to (3) at rate $\mu \ge \lambda$, uniformly in $\kappa$. The operator $P \mapsto FP + PF^\top$ has eigenvalues $\{-2\kappa, -(\lambda+\kappa), -2\lambda\}$, so a general initial condition relaxes as a combination of $e^{-2\kappa\rho}$ (the $\varepsilon$–$\varepsilon$ block), $e^{-(\lambda+\kappa)\rho}$ (the cross term) and $e^{-2\lambda\rho}$ (the $H$–$H$ block). Under (SS) the first has zero coefficient, because $P_{\varepsilon\varepsilon}$ is constant; the slower of the remaining two is $e^{-\mu\rho}$, which equals $e^{-2\lambda\rho}$ only when $\kappa \ge \lambda$.*

*Remark* 7 (the $\kappa$ floor). The first two bounds in (11) are uniform in $\kappa$; the third carries $\lambda/\kappa$ and is not. What it needs is weak: at $\rho = 1$ the deviation is $(\lambda/\kappa)e^{-2\lambda}$ to leading order, so relative accuracy $\epsilon$ needs $\kappa \gtrsim \lambda e^{-2\lambda}/\epsilon$, which at $\lambda = 10$, $\epsilon = 10^{-2}$ is $\kappa \geq 2 \times 10^{-6}$. Read through the correlation time: $\mathrm{Cov}(\varepsilon(t), \varepsilon(t+r)) \propto e^{-\kappa|r|}$ at stationarity, so $1/\kappa$ is the lag at which the maintenance noise decorrelates to $1/e$. A process with $1/\kappa$ of minutes is constant over the 3 s delay and over dozens of consecutive trials — a session-level drift, for which the model has no term. A floor $\kappa \geq 0.05\,\mathrm{s}^{-1}$ ($1/\kappa \leq 20$ s, about seven delay periods) is therefore a statement about what the term means, not a range imported from simulation.

# 4 Neglecting the transient

Let $\nu_{\mathrm{stat}}$ denote the rate computed from the stationary moments (3), i.e. with $\sigma_H^2 = A_\infty$, $\sigma_H^2 = B_\infty$, $c_H = 0$. This section establishes the single result the rest of the note rests on.

**Theorem 8** (the stationary approximation)**.** *Assume*

$$\text{(i) } \lambda \geq 5\,\mathrm{s}^{-1}, \qquad \text{(ii) } \kappa \geq 0.2\,\mathrm{s}^{-1}, \qquad \text{(iii) } a_\infty \leq 5. \tag{12}$$

*Then, throughout the analysis window $\rho \in [1, 3]$ s and for both models:*

(a) $\nu(\rho) = \nu_{\mathrm{stat}}(\rho)\, e^{\theta(\rho)}$ *with* $|\theta(\rho)| \leq \mathcal{E}(\rho)$ *and* $\mathcal{E}(1) \leq 0.36$ *— the rate is exact to within a factor* 1.43*, and to within* 1.0004 *by the end of the window;*

(b) *the least-squares slope and quadratic coefficient of* $\ln \nu$ *on the* 20 *bins of the window differ from those of* $\ln \nu_{\mathrm{stat}}$ *by at most*

$$\left|c_1 - c_1^{\mathrm{stat}}\right| \leq 0.081\ \mathrm{s}^{-1}, \qquad \left|c_2 - c_2^{\mathrm{stat}}\right| \leq 0.145\ \mathrm{s}^{-2};$$

(c) *under* $\lambda \geq 6$ *these become* $0.027\,\mathrm{s}^{-1}$ *and* $0.049\,\mathrm{s}^{-2}$*; under* $\lambda \geq 8$*,* 0.0030 *and* 0.0058*.*

The observed log-rate slope is $+0.867\,\mathrm{s}^{-1}$, so (b) is a 9% error bar on it and (c) a 3% one. *Everything from Section 5 on assumes* (12) *and works with* $\nu_{\mathrm{stat}}$*, carrying (b) as the error bar.* Sections 4.1 and 4.2 justify the hypotheses and prove the theorem.

## 4.1 The three conditions

Each can be argued without reference to the measured rate.

**(i)** $\lambda \geq 5\,\mathrm{s}^{-1}$**, i.e.** $1/\lambda \leq 200$ **ms.** $1/\lambda$ is the time constant of the low-pass filter carrying motor preparation to the gate. Dendauw et al. estimate $\lambda \approx 20\,\mathrm{s}^{-1}$ (50 ms) in an immediate-report task, so (12) is four times more permissive than the published value. $\lambda$ is in principle identifiable from these data, since it sets how fast the rate follows the stimulus; that check has not been carried out.

**(ii)** $\kappa \geq 0.2\,\mathrm{s}^{-1}$**, i.e.** $1/\kappa \leq 5$ **s.** $1/\kappa$ is the correlation time of the maintenance noise: $\mathrm{Cov}(\varepsilon(t), \varepsilon(t+r)) \propto e^{-\kappa|r|}$ at stationarity, so it is the lag at which the fluctuation has decorrelated to $1/e$. Requiring it to be at most one trial says that what is maintained fluctuates *within* a trial; a longer correlation time describes a drift carried across trials, for which the model has no term and which the randomised design averages out rather than expresses as a delay-period ramp. (i) and (ii) are substitutes: $\lambda \to \lambda + 1$ buys about what $\kappa \to 2\kappa$ buys, so only one needs to be argued hard.

**(iii)** $a_\infty \le 5$**.** Not fitted. By (19) the expected number of partial activations per 2 s delay window is $(\omega/\pi)e^{-a^2/2}$; at the largest $\omega$ in play this is one per 19 trials at $a = 3.3$, one per 246 at $a = 4$, and *one per* 22 000 *at* $a = 5$. So (iii) says only that the model must produce partial activations at all — the experiment comprises of order $10^4$ trials — not that it must reproduce the observed rate. Section 6 gives the sharper, data-based $a \le 3.3$, which is not used.

### 4.2 Proof of Theorem 8

**Proposition 9** (pointwise error)**.** *Fix* $\rho \ge 1$*, evaluate* $\bar\Delta, \bar\gamma, \bar e$ *of* (10) *at* $\rho$*, and write* $u_s := \beta/(\delta\sqrt{B_\infty})$*,* $c_g := \Phi(0)/g(0) = 1.2533$ *and*

$$\eta := \bar e + \frac{\bar\gamma^2}{\omega^2(1-\bar\Delta)}. \tag{13}$$

*If* $\bar\Delta < 1$ *and* $\eta < 1$ *then, for both models,*

$$\big|\ln\nu(\rho) - \ln\nu_{\mathrm{stat}}(\rho)\big| \;\le\; \mathcal{E}(\rho) := \frac{(a_\infty^2+1)\bar\Delta}{2(1-\bar\Delta)} + \frac{\eta}{2(1-\eta)} + c_g\left[\frac{\bar\gamma\, a_\infty}{\omega(1-\bar\Delta)\sqrt{1-\eta}} + u_s\left(\frac{1}{\sqrt{1-\eta}} - 1\right)\right]. \tag{14}$$

*Proof.* Write $\Delta := \sigma_H^2/A_\infty - 1$, $\gamma := c_H/A_\infty$, $E := \sigma_H^2/B_\infty - 1$, so $|\Delta| \le \bar\Delta$, $0 \le \gamma \le \bar\gamma$, $|E| \le \bar e$ by Proposition 5. From (16),

$$\ln\nu - \ln\nu_{\mathrm{stat}} = -\tfrac12\big(a^2 - a_\infty^2\big) - \tfrac12 \ln\frac{\sigma_H^2}{A_\infty} + \ln\frac{G}{G_s},$$

with $G = \sqrt{S}\, g(u)$ and $G_s = \sqrt{B_\infty}\, g(u_s)$. Throughout, $|\ln(1+x)| \le |x|/(1-|x|)$ for $|x| < 1$.

*(1)* $a^2 = a_\infty^2/(1+\Delta)$, so $\big|a^2 - a_\infty^2\big| \le a_\infty^2\bar\Delta/(1-\bar\Delta)$; halving and adding $\frac12|\ln(1+\Delta)| \le \bar\Delta/(2(1-\bar\Delta))$ gives the first term of (14).

*(2)* $S = B_\infty\big[(1+E) - \gamma^2/(\omega^2(1+\Delta))\big]$ since $A_\infty/B_\infty = \omega^{-2}$, so $|S/B_\infty - 1| \le \eta$, hence $S \ge B_\infty(1-\eta) > 0$ and $\frac12|\ln(S/B_\infty)| \le \eta/(2(1-\eta))$: the second term.

*(3)* $(\ln g)' = \Phi/g$ and $(\ln g)'' = -D/g^2 < 0$, where $D := \Phi^2 - \varphi g$ satisfies $D' = \varphi(\Phi + ug) > 0$ and $D(-\infty) = 0$, hence $D > 0$; so $\Phi/g$ is decreasing and $\sup_{u\ge0}(\ln g)' = \Phi(0)/g(0) = c_g$. Since $M = \beta/\delta + \gamma d/(1+\Delta) \ge \beta/\delta \ge 0$ we have $u, u_s \ge 0$, so $|\ln g(u) - \ln g(u_s)| \le c_g|u - u_s|$. Splitting $u - u_s = \frac{M-\beta/\delta}{\sqrt{S}} + \frac{\beta}{\delta}\big(\frac{1}{\sqrt{S}} - \frac{1}{\sqrt{B_\infty}}\big)$, the first piece is at most $\bar\gamma a_\infty/\big(\omega(1-\bar\Delta)\sqrt{1-\eta}\big)$ using $M - \beta/\delta = \gamma d/(1+\Delta)$, $d = a_\infty\sqrt{A_\infty}$ and $\sqrt{S} \ge \sqrt{B_\infty(1-\eta)}$; the second equals $u_s|(1+x)^{-1/2} - 1|$ with $|x| \le \eta$, hence at most $u_s(1/\sqrt{1-\eta} - 1)$. Multiplying by $c_g$ gives the third term. □

**Proposition 10** (transfer to the measured coefficients)**.** *On the* 20 *bin centres* $\rho_1, \dots, \rho_{20}$ *of the window, the least-squares slope and quadratic coefficient are linear functionals* $c_j = \sum_i w_i^{(j)} f_i$ *of the profile, with* $\sum_i |w_i^{(\mathrm{slope})}| = 1.504$ *and* $\sum_i |w_i^{(2)}| = 2.916$*. Hence*

$$\big|c_j - c_j^{\mathrm{stat}}\big| \;\le\; \sum_i \big|w_i^{(j)}\big|\,\mathcal{E}(\rho_i) \;\le\; \Big(\textstyle\sum_i |w_i^{(j)}|\Big)\,\mathcal{E}(1). \tag{15}$$

*The first form is the one used:* $\mathcal{E}$ *decays like* $e^{-\mu\rho}$ *across the window, so it is* 7 *to* 10 *times smaller than the uniform second form.*

*Proof.* Immediate from Proposition 9 and $\mathcal{E}$ decreasing (Proposition 5). □

*Proof of Theorem 8.* Under (12), $\mathcal{E}$ of (14) is increasing in $a_\infty$ and, at $\rho = 1$, maximised over $\kappa \ge 0.2$ at $\kappa = 0.2$; evaluating gives $\mathcal{E}(1) = 0.353$ at $\lambda = 5$, 0.133 at $\lambda = 6$ and 0.019 at $\lambda = 8$, which is (a). Feeding $\mathcal{E}(\rho_i)$ into (15) gives (b) and (c):

| $\lambda \geq$ | $1/\lambda$ | $\mathcal{E}(1)$ | slope error | curvature error |
|---|---|---|---|---|
| 4 | 250 ms | 0.96 | 0.253 | 0.446 |
| 5 | 200 ms | 0.353 | 0.081 | 0.145 |
| 6 | 167 ms | 0.133 | 0.027 | 0.049 |
| 8 | 125 ms | 0.019 | 0.0030 | 0.0058 |

□

*Remark* 11 (why the transfer step is needed). Proposition 5 bounds the *moments*; the observables are a slope and a curvature, that is, *derivatives*, and a bound on a function does not bound its derivative. Proposition 9 bridges the gap by bounding $\ln \nu$ itself, pointwise, after which (15) is elementary because $c_1$ and $c_2$ are bounded linear functionals of the profile.

## 5 The stationary rate

Let $Z := s_{\text{resp}} H - d$, so that partial activations are upcrossings of zero by $Z$ and $\dot{Z} = \dot{H} + \beta/\delta$. The Kac–Rice formula for a differentiable Gaussian process gives

$$\nu(t) = p_{Z(t)}(0)\, \mathbb{E}\big[\dot{Z}\mathbf{1}_{\{\dot{Z}>0\}} \mid Z = 0\big] = \frac{\varphi(a)}{\sigma_H}\, G, \qquad G = M\Phi(u) + \sqrt{S}\varphi(u) = \sqrt{S}\, g(u), \quad (16)$$

with $a = d/\sigma_H$, $M = \beta/\delta + (c_H/\sigma_H^2)d$, $S = \sigma_{\dot{H}}^2 - c_H^2/\sigma_H^2$ and $u = M/\sqrt{S}$.

**Proposition 12** (the stationary rate). *Substituting the stationary moments* (3) *($c_H = 0$) into* (16),

$$\nu_{\text{stat}} = \varphi(a)\left[\frac{\beta}{\delta\sigma_H}\Phi\left(\frac{\beta}{\delta\sigma_{\dot{H}}}\right) + \omega\,\varphi\left(\frac{\beta}{\delta\sigma_{\dot{H}}}\right)\right], \quad (17)$$

*whose bracket does not depend on d. Consequently:*

*(i)* ***Model 2*** *($\beta > 0$): $a(\rho) = a(0) - \beta\rho/(\delta\sigma_H)$ is affine, so exactly*

$$\frac{\mathrm{d}\ln\nu_{\text{stat}}}{\mathrm{d}\rho} = \frac{\beta}{\delta\sigma_H}\, a(\rho), \qquad \frac{\mathrm{d}^2\ln\nu_{\text{stat}}}{\mathrm{d}\rho^2} = -\left(\frac{\beta}{\delta\sigma_H}\right)^2, \quad (18)$$

*a downward parabola, the same one in all six conditions.*

*(ii)* ***Model 1*** *($\beta = 0$): $M = 0$, $u = 0$, a constant, and*

$$\nu_{\text{stat}} = \frac{\omega}{2\pi}\, e^{-a^2/2}, \qquad \frac{\mathrm{d}\ln\nu_{\text{stat}}}{\mathrm{d}\rho} \equiv 0, \qquad \frac{\mathrm{d}^2\ln\nu_{\text{stat}}}{\mathrm{d}\rho^2} \equiv 0. \quad (19)$$

*Proof.* Set $c_H = 0$ in (16); then $\ln\nu_{\text{stat}} = \text{const} - a^2/2$ and (18) follows from $a$ affine. For (ii) set $\beta = 0$: $M = 0$, $g(0) = 1/\sqrt{2\pi}$, $G = \sqrt{B_\infty}/\sqrt{2\pi} = \omega\sigma_H/\sqrt{2\pi}$, so $\nu_{\text{stat}} = \varphi(a)\omega/\sqrt{2\pi} = \omega e^{-a^2/2}/2\pi$, constant in $\rho$. □

**Proposition 13** (level orders by distance). From (17). *Under both models $\nu_{\text{stat}} = \varphi(a) \times$ (a factor independent of d), so $\nu_{\text{stat}}$ is strictly decreasing in a, with no condition whatever. Under Model 2 the slope $\beta a/(\delta\sigma_H)$ of* (18) *is strictly* increasing *in a. Hence level and slope order oppositely across conditions.*

*Remark* 14 (before the window). The same monotonicity holds during the transient, though it then needs a condition: from (16), $\partial\ln\nu/\partial d = -a/\sigma_H + c_H\Phi(u)/(\sigma_H^2 G)$ is negative iff $dG > c_H\Phi(u)$, and $G > M\Phi(u)$ with $dM = (\beta/\delta)d + c_H a^2 \geq c_H$ whenever $a \geq 1$. Not needed for anything below.

## 6 The standardised distance is measured

Inverting (19) — Model 1's own stationary rate formula, so using it to bound Model 1 is legitimate —

$$a = \sqrt{2\ln\Big(\frac{\omega}{2\pi\nu}\Big)}. \tag{20}$$

(C3) uses only the weak, data-free reading of this identity — $a \le 5$, on pain of the model producing fewer than one partial activation in twenty thousand trials. Two sharper readings are available if one is willing to use the measured rate, and neither is needed above. First, with the observed delay-period rates 0.030–0.135 $\mathrm{s}^{-1}$ and $(\lambda, \kappa)$ anywhere in the sampled range, (20) gives $a \le 3.3$, which tightens every ceiling in Section 4.1 by about a third. Second, (20) is the rate–ramp trade-off in its sharpest form: the rate *measures* the standardised distance, so the ramp Model 1 could produce and the rate it must produce are not free of one another.

## 7 Predictions

The delay-period observations are: **(L1)** the level rises with duration; **(L2)** the level rises with coherence; **(L3)** the coherence effect on the level is larger after the long stimulus; **(S1)** the log-rate slope is $+0.867\,\mathrm{s}^{-1}$; **(S2)** it is shallower after the long stimulus, $0.896 \to 0.723$; **(S3)** shallower at high coherence, $0.871 \to 0.726$. Throughout we assume (12) and use $\nu_{\mathrm{stat}}$, with the error bars of Theorem 8(b): $\pm 0.081\,\mathrm{s}^{-1}$ on a slope at $\lambda \ge 5$, $\pm 0.027$ at $\lambda \ge 6$.

**Corollary 15** (L1, L2: the level)**.** From Proposition 13, plus the saturation argument of the main text, which concerns the distribution of $m$ and is untouched by (ss). *Duration and coherence each raise the fraction of trials whose maintained target has reached the cap $\theta$, hence lower $d$ in the sense of first-order stochastic dominance; since $\nu_{\mathrm{stat}}$ is decreasing in $d$ at every $\rho$, and averaging over trials preserves the ordering, both raise the level. Predicted by* both *models:* not discriminating.

**Corollary 16** (L3: the interaction)**.** From Proposition 13 and the same saturation argument. *As the capped fraction tends to one the coherence levels converge, and duration raises that fraction at every coherence, so the coherence effect on the level must* shrink *with duration. Predicted by both models, and* contradicted *by the data. Not discriminating; a genuine disagreement.*

**Corollary 17** (S1: the slope)**.** From Proposition 12(ii) and Theorem 8(b). *Model 1 predicts a slope of exactly* 0*, hence a measured slope of at most* $0.081\,\mathrm{s}^{-1}$ *(at $\lambda \ge 5$;* 0.027 *at $\lambda \ge 6$). Model 2 predicts $\beta a(\rho)/(\delta\sigma_H) > 0$, its only source of a ramp.* Discriminating: the observed $+0.867\,\mathrm{s}^{-1}$ exceeds Model 1's ceiling by a factor of 10.8 at $\lambda \ge 5$ and 33 at $\lambda \ge 6$.

**Corollary 18** (S2, S3: the slope across conditions)**.** From Propositions 12, 13 and 15, with Theorem 8(b). *Model 1 predicts a slope of zero in* every *condition, hence a condition difference of at most twice the slope error:* $0.161\,\mathrm{s}^{-1}$ *at $\lambda \ge 5$,* 0.053 *at $\lambda \ge 6$. Model 2 predicts a difference of definite sign: the slope increases in a while the level decreases in a, so any manipulation that raises the level lowers the slope, and by Corollary 15 duration and coherence both raise the level.* The observed differences 0.173 and 0.145 carry the sign Model 2 requires; at $\lambda \ge 6$ they exceed Model 1's ceiling by 3.3 and 2.7, at $\lambda \ge 5$ only marginally. This is the leg that pays for a low $\lambda$; S1 does not.

*Remark* 19 (on the curvature). Under the cold start the curvature of the log rate discriminated, through the decay that Model 1's variance filling forces. Under (ss) there is no decay to force: both models predict a near-zero curvature (Model 1 exactly 0, Model 2 $-\frac{1}{2}(\beta/\delta\sigma_H)^2$, which is $\ge -0.034$ at the observed slope with $a \le 3.3$), and the observed $-0.0405 \pm 0.0416$ is consistent with both. The curvature therefore carries nothing here and is dropped from the list of tested observations; the discriminating role passes to S2 and S3.

## 8 Summary

Under (12) — $\lambda \geq 5$, $\kappa \geq 0.2$, $a_\infty \leq 5$ — with the error bars of Theorem 8:

| | Model 1 | Model 2 | verdict |
|---|---|---|---|
| L1, L2 level | rises | rises | shared |
| L3 interaction | shrinks | shrinks | shared; both fail |
| S1 slope | $0 \pm 0.081$ | $\beta a/\delta\sigma_H > 0$ | **discriminating**, 10.8× |
| S2, S3 by condition | $0 \pm 0.161$ | sign forced | sign; magnitude at $\lambda \geq 6$ |

At $\lambda \geq 6$ the ceilings are 0.027 and 0.053, and all three of S1–S3 discriminate by magnitude (33×, 3.3×, 2.7×).

## 9 Numerical verification

| Claim | Test | Result |
|---|---|---|
| Prop. 2 | vs. numerical integration of $\dot{P} = FP + PF^\top + Q$ | max rel. err. $8.9 \times 10^{-12}$ |
| Prop. 2 | the three identities in terms of $X$ | agree to $1.5 \times 10^{-13}$ |
| Prop. 5 | $0 \leq -X \leq (\lambda+\kappa)\rho e^{-\mu\rho}$ (50-digit) | worst ratio 1.000000 |
| Prop. 5 | $\lvert\sigma_H^2/A_\infty - 1\rvert \leq \bar{\Delta}$ | worst ratio 1.000000 |
| Prop. 5 | $c_H/A_\infty \leq \bar{\gamma}$ | worst ratio 1.000000 |
| Prop. 5 | $\lvert\sigma_H^2/B_\infty - 1\rvert \leq (\lambda/\kappa)\bar{\Delta}$ | worst ratio 0.997226 |
| Prop. 5 | tightened $\bar{\Delta}, \bar{\gamma}, \bar{e}$ (50-digit) | worst ratios 1.000000 (all three) |
| Prop. 9 | $\lvert\ln\nu - \ln\nu_{\text{stat}}\rvert \leq \mathcal{E}$, 71 079 points | 0 violations, worst ratio 0.612 |
| Prop. 10 | the two functional norms 1.504, 2.916 | computed exactly |

Grids: $\lambda \in [0.3, 60]$, $\kappa \in [0.01, 60]$, $\rho \in [0.05, 6]$, $\beta/\delta \in \{0, 0.05, 0.3, 1, 3\}$, $a_\infty \in \{0.5, 1, 2, 3.3, 6, 10\}$. The $X$ bounds were verified in 50-digit arithmetic because the combination $P_{\varepsilon\varepsilon} - 2P_{\varepsilon H} + P_{HH}$ cancels catastrophically in double precision; points where $\mathcal{E}$ falls below double-precision resolution ($10^{-10}$) were excluded from the Theorem 9 check for the same reason.

## 10 Consequences for the paper

Adopting (ss) removes from the appendix the convolution kernel $K$, the integrals $\int K^2$ and $\int K'^2$, the auxiliary $W = K' - K^3/2\sigma_H^2$, the log-concavity lemma, the non-stationary derivative of $\ln\nu$, and the transient content of the present Corollaries A.2 and A.3. What remains is Sections 2–4 — five short results establishing that the transient may be neglected — followed by the stationary rate and the predictions as one-line corollaries.

The cost is that theory and simulation must agree, so the 1200-draw parameter sweep, the admissibility screen and every effect size would have to be recomputed under (ss).